FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Advanced)-2019

PAPER – 1

TEST DATE: 05-05-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B, D

Sol. $\left(\vec{A}\times\vec{B}\right)\vec{C}-\left(\vec{A}\times\vec{C}\right)\times B=0$

$\left(\vec{C}.\vec{B}\right)\vec{A}-\left(\vec{C}\vec{A}\right)\vec{B}-\left(\vec{B}.\vec{C}\right)\vec{A}+\left(\vec{B}.\vec{A}\right)=0$

$\left(\vec{B}.\vec{A}\right)\vec{C}-\left(\vec{C}\vec{A}\right)\vec{B}=0$

$\vec{A}\times\left(\vec{B}\times\vec{C}\right)=0$

2. A, D

Sol. $U_y=V_0\sin\alpha$

$U_x=V_0+V_0\cos\alpha$

$\frac{U_y}{U_x}=\tan\frac{\alpha}{2}$

When throwing point and two stones are collinear

$\frac{y_1}{x_1}=\frac{y_2}{x_2}\Rightarrow t=\frac{2V_0}{g}\tan\frac{\alpha}{2}$ (where top point of pole is origin and x_1, x_2, y_1, y_2 are measured from there)

and $h<\frac{1}{2}gt^2$

3. A, C

Sol. $\tau = (kx)R = kR^2\theta$

$I\alpha = kR^2\theta$

$\left(\frac{MR^2}{2} + mR^2\right)\alpha = kR^2\theta$

$T = 2\pi\sqrt{\frac{\frac{M}{2}+m}{k}}$

4. A, D

Sol. Basic concept of charging and discharging.

5. B

Sol. Shape of conductor does not make any difference.

6. A, B, D

Sol. Theory of wave optics

7. B

8. B

Sol. for (7-8)

$v = \frac{660}{31}$ m/s

If duration of beep is t

then

$\left(\frac{330t}{v+330}\right) = \frac{100}{v}$

9. B

10. D

Sol. for (9 – 10)

Total charge in an element = $Rd\theta \times \lambda_0 \sin\theta$

On revolving this ring we get a sphere of charge density = $\frac{Rd\theta\lambda_0 \sin\theta}{2\pi R \sin\theta Rd\theta}$ which is a constant.

Basic concept of potential

$V_A - V_B = E.R$

SECTION – D

11. 00060.00

Sol. $\sqrt{a^2 + a^2 + 2a^2\cos(180-\alpha)} = a$

$\alpha = 60°$

12. 00012.50

Sol. For equilibrium centre of mass should be on the wire.

13. 00025.02

Sol. $\frac{dh}{dt} = \frac{\sqrt{2g(h-76)}}{\sqrt{n^2-1}}$

$$\int_{0.96}^{0.76} \frac{dh}{\sqrt{h-76}} = \int_{0}^{5} \frac{\sqrt{2g}\ dt}{\sqrt{n^2-1}}$$

14. 00015.00

Sol. Calculate by potential method.

15. 00004.00

Sol. $f_{Beat} = f_1 - f_2$

$$= f\left(\frac{c}{c-v}\right) - f\left(\frac{c}{c+v}\right)$$

$$f_{beat} = \frac{2fv}{c}$$

16. 00017.50

Sol. Object must be at 2f distance of combination

$$\frac{1}{f_{net}} = \frac{-2}{f_L} + \frac{1}{f_M}$$

17. 00001.00

Sol. $F = \frac{-edV}{dr} = -\frac{eV_0 r_0}{r}$

$$\frac{mv^2}{r} = \frac{eV_0 r_0}{r}$$

v = constant

$$mvr = \frac{nh}{2\pi}$$

$r \propto n$

$$M = \frac{evr}{2} \Rightarrow M \propto n$$

18. 00001.50

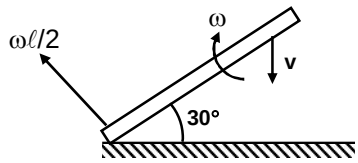
Sol. Constraint at contract point

$$\frac{\omega\ell}{2}\cos 30° = v$$

energy conservation

$$mg\frac{\ell}{4} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$v = \sqrt{\frac{g}{26}g\ell}$$



Chemistry

PART – II

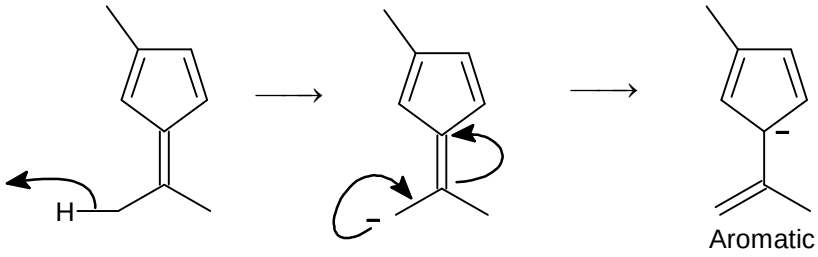
SECTION – A

19. C, D

Sol. As per Usanovich concept, in the first reaction As_2S_3 will behave as a acid and $(NH_4)_2S$ will behave as a base. Similarly in second reaction $Fe(CN)_2$ is acid and KCN is base.

20. B

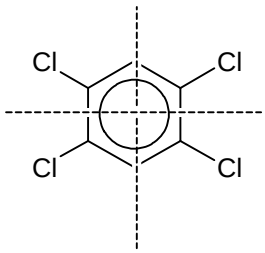
Sol.



Aromatic

21. A

Sol.



xy plane
yz plane
zx plane

22. C

Sol. Equilibrium constant for the formation of a hydrate $\propto$ electron deficient nature of the carbon of carbonyl group (C = O). In the fourth compound the carbon of carbonyl group will be more electron deficient.

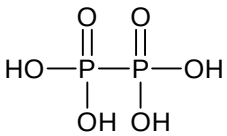
23. A, B, C, D

Sol. Properties of orthoboric acid.

$$H_3BO_3 \xrightarrow{\Delta} B_2O_3$$

24. A, C

Sol.



Hypophosphoric acid

It is tetrabasic
No P–O–P linkage
P atoms show maximum covalency
O. S. of P is 4

25. C

26. A

Sol. $FeCl_3 \rightarrow$ Yellow solution and does not react with dil H_2SO_4

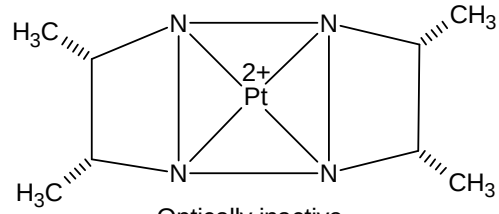
$$2Fe^{3+} + H_2S \xrightarrow{H^+} 2\underset{U}{Fe^{2+}} + S$$

$$Fe^{3+} + 3NaOH \underset{(excess)}{\longrightarrow} \underset{\text{Brown ppt (V)}}{\left(Fe(OH)_3\right)\downarrow} + 3Na^+$$

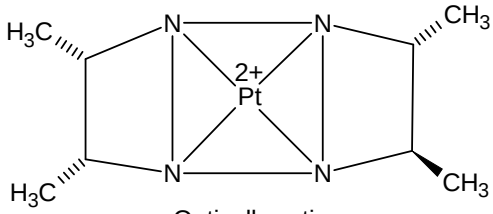
27. B

28. D

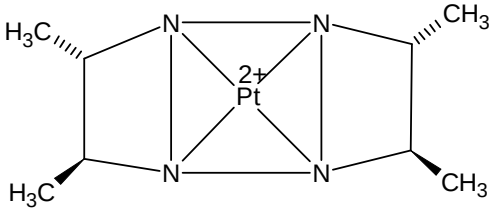
Sol.



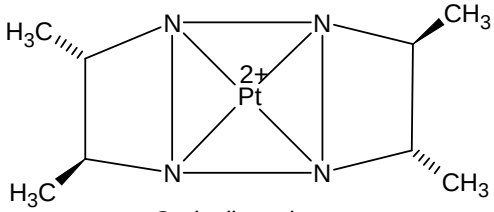
Optically inactive



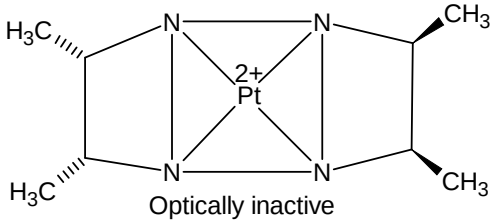
Optically active



Optically inactive



Optically active



Optically inactive

SECTION – D

29. 00002.22

Sol. $N_2 \rightleftharpoons 2N$

$\frac{2.8}{28}$ 0

$\frac{2.8}{28} \times \frac{50}{100}$ $\frac{2.8}{28} \times 2 \times \frac{50}{100}$

Total moles = $\frac{3}{20}$

PV = nRT

$P \times 10 = \frac{3}{20} \times 0.082 \times 1805$

P = 2.22

30. 00036.36

Sol. $Fe + Fe_2(SO_4)_3 \rightarrow 3FeSO_4$

1.25 gm ferric alum produces 0.5913 gm $FeSO_4$

Meq of $FeSO_4$ = Meq of $KMnO_4$

$$\frac{0.5913}{152}\times 1000 = 0.107\times V$$

V = 36.36 ml

31. 00001.25

Sol. There are four co-ordination isomers are possible for the given complex. These are

$$\left[Cu(NH_3)_4\right][PtCl_4], \left[Cu(NH_3)_3Cl\right][PtCl_3NH_3],$$

$$\left[Cu(NH_3)Cl_3\right]\left[Pt(NH_3)_3Cl\right], \left[CuCl_4\right]\left[Pt(NH_3)_4\right]$$

32. 00009.50

Sol. $HA \rightleftharpoons H^+ + A^-$

$$\left[H^+\right] = \sqrt{CK_a}$$

$Ka = 10^{-8}$

$BOH \rightleftharpoons B^+ + OH^-$

$$\left[OH^-\right] = \sqrt{CK_b}\,(K_a = K_b)$$

$= \sqrt{10^{-9}}$

$$\left[H^+\right] = \frac{10^{-14}}{10^{-\frac{9}{2}}}$$

= pH = 9.50

33. 00001.25

Sol. As we know,

$$t_{1/2} \propto a_0^{1-n}$$

$$\Rightarrow t_{1/2} = c.(a_0)^{1-n}\left[c = \text{constant}\right]$$

$$\log t_{1/2} = \log c + (1-n)\log a_0$$

$$\therefore [1-n] = \tan 45^\circ \Rightarrow 1-n = 1 \Rightarrow n = 0$$

Now, for K,

As we know for zero order reaction,

$$t_{1/2} = \frac{a_0}{2K}$$

$$\Rightarrow \log t_{1/2} = \log\left(\frac{a_0}{2K}\right) = \log a_0 + \log\frac{1}{2K} \qquad \ldots (1)$$

From graph,

$$\log\frac{1}{2K} = 2$$

$$K = \frac{1}{200}\,\text{mol}\,l^{-1}\,\text{sec}^{-1}$$

$$\therefore 250\ K = 250\times\frac{1}{200} = 1.25\ \text{mol}/l/\text{sec}$$

34. 00155.88

Sol. $$\frac{a\sqrt{3}}{4} = 2r$$

2r = 155.88 pm

35. 00003.63

Sol. For a polyprotic process:

$PV^x = K \left[x \notin \{1, \gamma\}\right]$

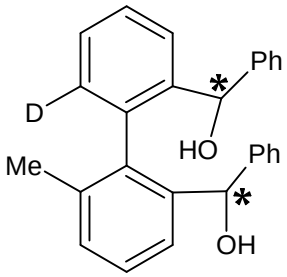
$$C = C_{V,m} + \frac{R}{1-x}$$

$$C = \frac{3}{2}R + \frac{R}{3}$$

$$= \frac{11R}{6} = \frac{11 \times 1.98}{6} = 3.63$$

36. 00001.60

Sol. P is:



Total number of stereo isomers in P is 8.

$$\therefore \frac{x}{5} = \frac{8}{5} = 1.60$$

Mathematics PART – III

SECTION – A

37. C, D

Sol. $L = \lim_{x\to 0} \frac{ae^x + b\ln(1+x) + 5}{x^2} \Rightarrow L = \lim_{x\to 0} \frac{a\left(1+\frac{x}{1!}+\frac{x^2}{2!}+.....\right)+b\left(x-\frac{x^2}{2}+\frac{x^3}{3}.....\right)+5}{x^2}$

$\Rightarrow a = -5, b = 5, L = -5$

38. A, B, C

Sol. Let $\lim_{n\to\infty}\prod_{k=3}^{n}\left(1-\tan^4\frac{\pi}{2^k}\right) = L \Rightarrow L = \lim_{n\to\infty}\cos\frac{\pi}{4}\prod_{k=3}^{n}\left(\cos\left(\frac{\pi}{2^k}\right)\right)^{-3}$

As $\frac{\sin x}{x} = \prod_{k=1}^{\infty}\cos\left(\frac{x}{2^k}\right) \Rightarrow \frac{\sin x}{x\cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)} = \prod_{k=3}^{\infty}\cos\left(\frac{x}{2^k}\right)$

$\Rightarrow \lim_{x\to\pi}\frac{\sin x}{x\cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)} = \lim_{x\to\pi}\prod_{k=3}^{\infty}\cos\left(\frac{x}{2^k}\right) \Rightarrow \prod_{k=3}^{\infty}\cos\left(\frac{x}{2^k}\right) = \frac{2\sqrt{2}}{\pi} \Rightarrow L = \frac{1}{\sqrt{2}}\left(\frac{\pi}{2\sqrt{2}}\right)^3 = \frac{\pi^3}{32}$

39. A, B

Sol. Let $a_1 = a$ and $b_2 - a_1 = d \Rightarrow (a + 4d)^2 = (a + 2d)(a + 7d) \Rightarrow a = 2d$

Also, $a_9 = 40 \Rightarrow a + 8d = 40 \Rightarrow a = 8, b = 4 \Rightarrow \sum_{i=1}^{9} a_i^2 = 6144$

Also, $b_1 = 54, \frac{b_2}{b_1} = \frac{3}{2} \Rightarrow \sum_{i=1}^{\infty}\frac{1}{b_i} = \frac{1}{18}$

40. B, D

Sol. A = Coefficient of x^{m-2} in $(1+x)^m(x+1)^m - {}^mC_0 \cdot {}^mC_2 \Rightarrow A = {}^{2m}C_{m-2} - {}^mC_2$

41. A, B, C, D

Sol. Exponent of '2' is same in 15!, 14!·1!, 13!·2!, ...

$\Rightarrow$ Number of values of 'r' = 16 $\Rightarrow$ k = 16

42. A, C, D

Sol. $\text{Arg}\left(\frac{z+z_1}{z_2-z_3}\right) = 0$ or π

43. D

Sol. $\int_0^x\left(a_0 + a_1x + a_2x^2 + ... + a_{125}x^{125}\right)dx = \int_0^x\left(x^3+x^5\right)\left(2x^6+3x^4-1\right)^{20}dx$

$\Rightarrow a_0x + a_1\frac{x^2}{2} + a_2\frac{x^3}{3} + a_3\frac{x^4}{4} + ... + a_{125}\frac{x^{126}}{126} = \frac{\left(2x^6+3x^4-1\right)^{21}+1}{252}$

In the expansion of $\frac{\left(2x^6+3x^4-1\right)^{21}+1}{252}$, we have only even powers of 'x'

$\Rightarrow a_0 = a_2 = a_4 = ... = a_{124} = 0$

44. D

Sol. $a_0 = a_2 = a_4 = ... = a_{124} = 0$
Also, $a_1 = 0$ and $a_{125} = 2^{20}$

45. C
46. B
Sol. (for Q.45.-46)

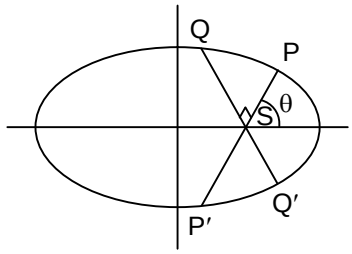
$$I\left(\frac{1}{\ln 2}\right) = I(\log_2 e) \Rightarrow I(\log_2 e) = \int_0^{2\pi} \frac{2^{\log_2 e}}{2^{\log_2 e} + 2^{-\log_2 e}} dx \Rightarrow I(\log_2 e) = \int_0^{2\pi} \frac{e}{e + \frac{1}{e}} dx$$

$$\Rightarrow I(\log_2 e) = \frac{2\pi e^2}{e^2 + 1} \Rightarrow I(\sin x) = \int_0^{2\pi} \frac{2^{\sin x}}{2^{\sin x} + 2^{-\sin x}} dx \Rightarrow I(\sin x) = \int_0^{2\pi} \frac{2^{-\sin x}}{2^{-\sin x} + 2^{\sin x}} dx$$

Adding both
$\Rightarrow 2I(\sin x) = 2\pi \Rightarrow I(\sin x) = \pi$

SECTION – D

47. 00004.00
Sol. $P \equiv (ae + r\cos\theta, r\sin\theta)$



$$\frac{(\sqrt{10}e + r\cos\theta)^2}{10} + \frac{(r\sin\theta)^2}{6} = 1$$

$$r^2\left(\frac{\cos^2\theta}{10} + \frac{\sin^2\theta}{6}\right) + \frac{10e^2 - 10}{10} + \frac{2\sqrt{10}er\cos\theta}{10} = 0$$

r_1 r_2

$$r_1 r_2 = \frac{(10e^2 - 10)60}{10 \cdot (6\cos^2\theta + 10\sin^2\theta)}$$

$$|SP \cdot SP'| = \frac{(1 - e^2)60}{6\cos^2\theta + 10\sin^2\theta} \Rightarrow \frac{1 - e^2}{SP \cdot SP'} = \frac{6\cos^2\theta + 10\sin^2\theta}{60}$$

Similarly for Q coordinate replace θ by $90^\circ - \theta$

It gives $\frac{10\cos^2\theta + 6\sin^2\theta}{60} = \frac{1 - e^2}{SQ \cdot SQ'}$

$$\therefore \frac{1 - e^2}{SQ \cdot SQ'} + \frac{1 - e^2}{SQ \cdot SQ'} = \frac{6\cos^2\theta + 10\sin^2\theta}{60} + \frac{10\cos^2\theta + 6\sin^2\theta}{60} = \frac{16}{60} = \frac{4}{15} = \frac{N}{15}$$

$\therefore N = 4$

48. 00005.00

Sol. $$P = \prod_{k=1}^{n} \frac{4(2k-1)^4 + 1}{4(2k^4) + 1}$$

$4a^4 + b^4 = ((a + b)^2 + a^2)((a - b)^2 + a^2)$

$$P = \prod_{k=1}^{n} \frac{\left(4k^2 + (2k-1)^2\right) \cdot \left((2k-2)^2 + (2k-1)^2\right)}{\left((2k)^2 + (2k+1)^2\right) \cdot \left(4k^2 + (2k-1)^2\right)}$$

$$P = \prod_{k=1}^{n} \frac{(2k-2)^2 + (2k-1)^2}{(2k)^2 + (2k+1)^2}$$

$$P = \frac{0^2 + 1^2}{2^2 + 3^2} \cdot \frac{2^2 + 3^2}{4^2 + 5^2} \cdot \frac{4^2 + 5^2}{6^2 + 7^2} ..... \frac{(2n-2)^2 + (2n-1)^2}{(2n)^2 + (2n+1)^2}$$

$$P = \frac{1}{4n^2 + (2n+1)^2} = \frac{1}{8n^2 + 4n + 1} \Rightarrow k_1 = 8, k_2 = 4, k_3 = 1$$

$\therefore k_1 + k_3 - k_2 = 5$

49. 00001.00

Sol. $(56x+33y)i=\frac{-iy}{x^2+y^2},\ 33x-56y=\frac{x}{x^2+y^2}$ $(z = x + iy)$

$56iz+33z=\frac{x-iy}{x^2+y^2}$

$56iz+33z=\frac{1}{z}\Rightarrow z^2=\frac{1}{33+56i}\Rightarrow z=\pm\frac{1}{7+4i}=\pm\frac{7-4i}{65}$

$|x|+|y|=\frac{11}{65}=\frac{p}{q}\Rightarrow 6p-q=66-65=1$

50. 00041.00

Sol. $\Rightarrow x^2+\frac{1}{x^2}+b+a\left(x+\frac{1}{x}\right)=0\Rightarrow x+\frac{1}{x}=t$

$t^2 + at + b - 2 = 0 \Rightarrow at + b + t^2 - 2 = 0,\ t^2 \in [4, \infty)$

This represents equation of line in a-b plane and $a^2 + b^2$ represent square of distance of a point on this line from O (origin)

$d=\frac{t^2-2}{\sqrt{1+t^2}}\Rightarrow t^2\in[4,\infty),\ d_{min}=\frac{2}{\sqrt{5}}$ at $t^2 = 4$

$d_{min}^2=\frac{4}{5}=\frac{p}{q}\Rightarrow p^2+q^2=41$

51. 00010.00

Sol. $P(\text{tail}) = \frac{n}{2n+1}\cdot 1+\frac{n+1}{2n+1}\cdot\frac{1}{2}=\frac{31}{42}\Rightarrow n = 10$

52. 00008.00

Sol. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$

$\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

$\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$

$\vec{a}\times\vec{b}=D_1\hat{i}+D_2\hat{j}+D_3\hat{k}$

$|\vec{a}\times\vec{b}|=\sqrt{D_1^2+D_2^2+D_3^2}=4$

$|c|=\sqrt{c_1^2+c_2^2+c_3^2}=2$

Maximum value of $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=|\vec{a}\times\vec{b}|\,|\vec{c}|=4\times 2=8$

53. 00001.00

Sol. $\int_1^{\sqrt{2}}\left((x^4-2x^2+1)^{\frac{1}{3}}-1\right)dx+\int_{-1}^{0}\sqrt{(x+1)^{\frac{3}{2}}+1}\,dx=1$

$\Rightarrow \int_1^{\sqrt{2}}(x^4-2x^2+1)^{\frac{1}{3}}dx+\int_{-1}^{0}\sqrt{(x+1)^{\frac{3}{2}}+1}\,dx=1+\sqrt{2}-1\Rightarrow I=\sqrt{2}$

54. 00004.00

Sol. Put $e^x = t \Rightarrow e^x\,dx = dt$

$\Rightarrow \int\frac{2e^{5x}+e^{4x}-4e^{3x}+4e^{2x}+2e^x}{(e^{2x}+4)(e^{2x}-1)^2}dx=\tan^{-1}\left(\frac{e^x}{2}\right)-\frac{1}{2(e^{2x}-1)}+c$

$\Rightarrow a + b = 4$